SPECIFICATION FOR APPROVAL

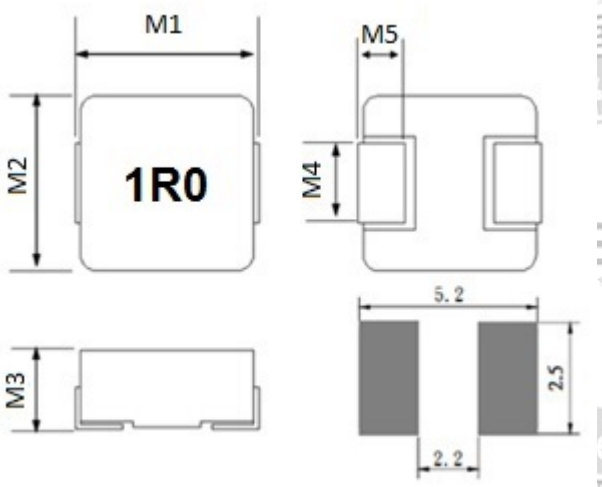
※This is a RoHS and REACH compliant product whose related documents are available on request.
※Graphic is only for dimensionally application.

1. PART NUMBERING IDENTIFICATION

MCS 0420- □□□ □ □□

Control	(Control No:)
Tolerance	(M=±20%)
Inductance	(1R0=1.0μH)
Dimensions	(0420=4.7 x 4.3 x 2.0mm)
Product symbol	(Series No:)

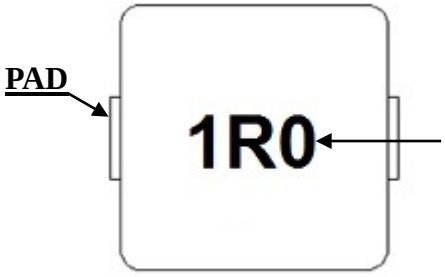
2. MECHANICAL DIMENSION



UNIT: mm

	DIM.	TOL.
M1	4.45	±0.25
M2	4.0	±0.3
M3	1.8	±0.2
M4	1.5	±0.3
M5	0.8	±0.3

3. MARKING



Marking Direction: PAD on the left and right sides, font facing up.
Example: 1R0 Stands for Marking➔1.0μH

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4. ELECTRICAL SPECIFICATION

Part number	Inductance (uH) ±20%	DC Resistance (mΩ) Typical	DC Resistance (mΩ) MAX.	Rated Current (A) Typical	I sat (A) Typical
MCS0420-R10MN2	0.10	3.5	4.0	12.0	22.0
MCS0420-R12MN2	0.12	4.5	5.0	12.0	12.0
MCS0420-R15MN2	0.15	6.0	6.6	9.0	13.0
MCS0420-R22MN2	0.22	6.0	6.6	9.0	12.5
MCS0420-R33MN2	0.33	9.6	11.0	10.0	13.0
MCS0420-R47MN2	0.47	12.5	14.0	7.0	9.5
MCS0420-R56MN2	0.56	14.0	16.0	6.5	10.0
MCS0420-R68MN2	0.68	16.0	18.0	6.0	9.0
MCS0420-1R0MN2	1.0	24.0	27.0	4.5	7.0
MCS0420-1R2MN2	1.2	24.0	27.0	4.5	7.0
MCS0420-1R5MN2	1.5	38.0	46.0	4.0	6.0
MCS0420-2R2MN2	2.2	52.0	58.0	3.0	5.0
MCS0420-3R3MN2	3.3	74.0	87.0	2.5	4.0
MCS0420-4R7MN2	4.7	98.0	110.0	2.2	3.5
MCS0420-5R6MN2	5.6	105.0	115.0	1.8	3.5
MCS0420-6R8MN2	6.8	160.0	175.0	1.5	2.5
MCS0420-100MN1	10.0	256.0	282.0	1.2	2.2

TEST INSTRUMENT: CHROMA 16502、Zentech1320+Zentech3305

NOTE:

1. **Test Freq.: 100KHz, 1.0V**
2. **All test data is referenced to 25℃ ambient.**
3. **Operating Temperature Range -25℃~+125℃.**
4. **Storage Temperature Range: -20℃~+40℃(<60% R.H.).**
5. **Rated Current: DC current(A)that will cause an approximate △T of 40℃.**
6. **I sat: DC current (A) that will cause Lo to drop approximately 30%.**
7. **The part temperature(ambient +temp rise)should not exceed 125℃ under worst case operating conditions.**
8. **Circuit design, component placement, PWB trace size and thickness, airflow and other cooling provisions all affect the part temperature Part temperature should be verified.**
9. **MSL: Level 1**

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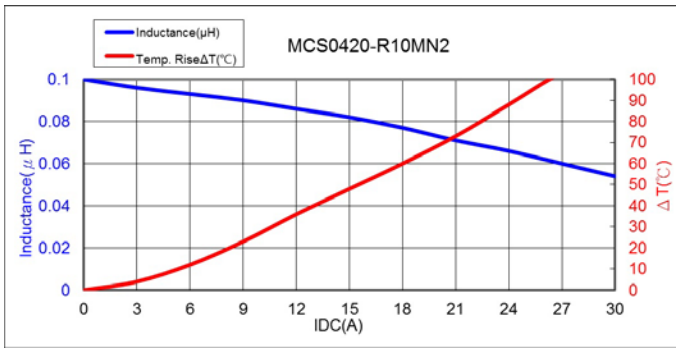
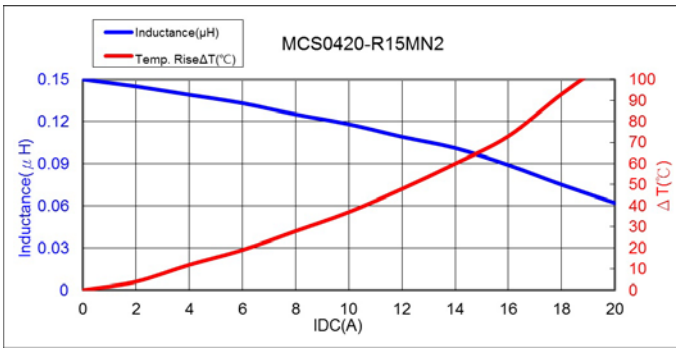
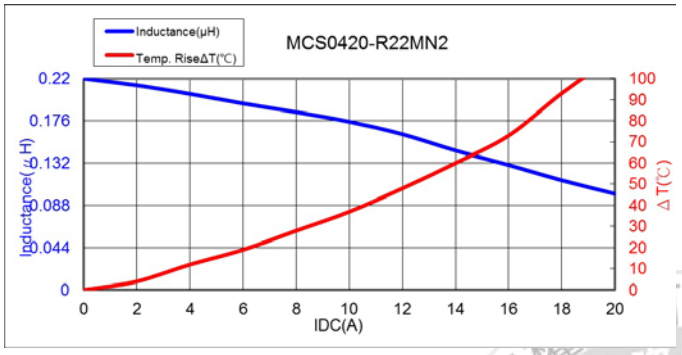
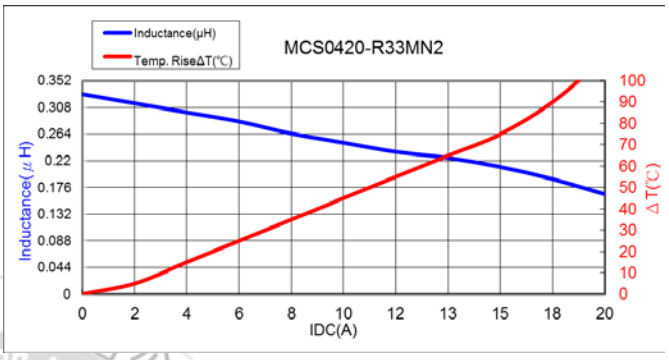
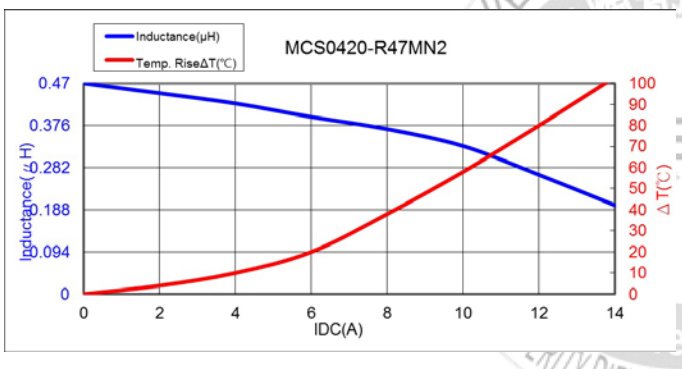
5. ELECTRICAL CURVE

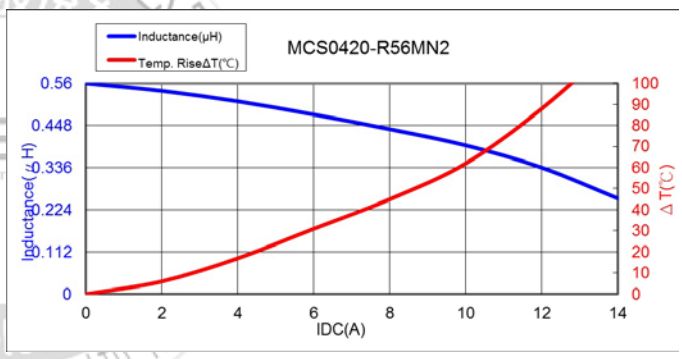


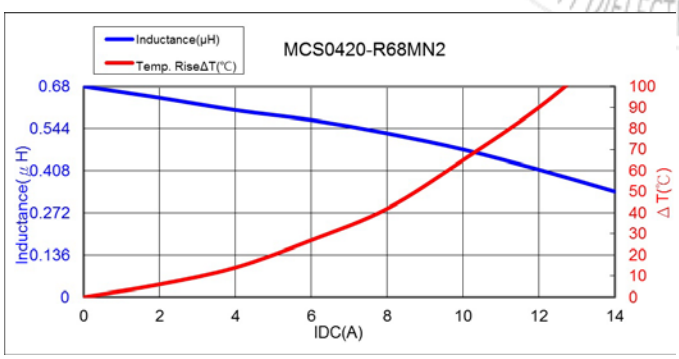


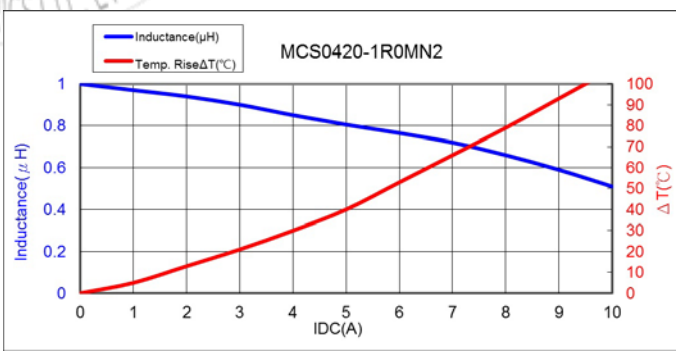


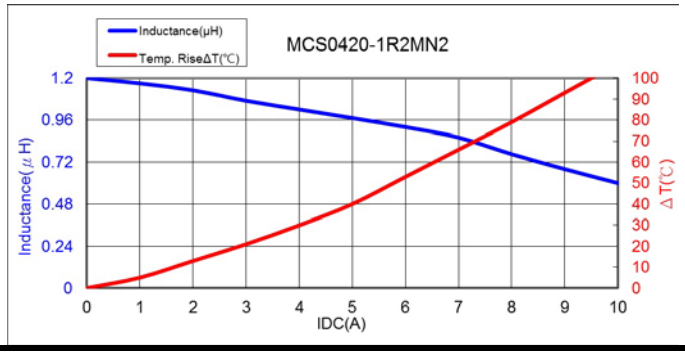


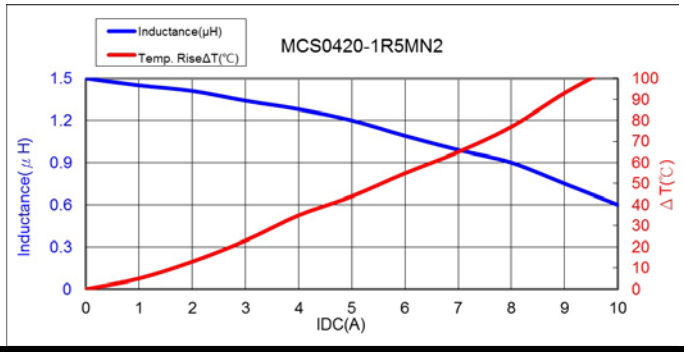
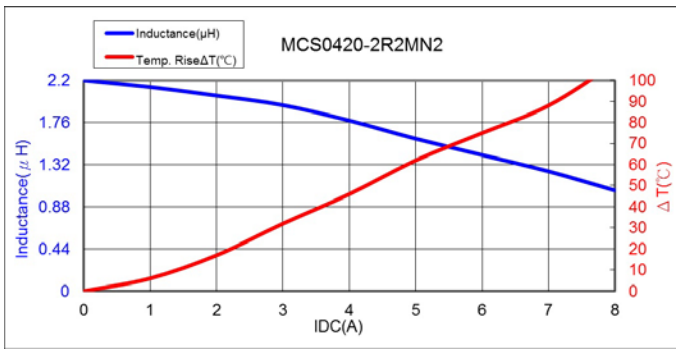
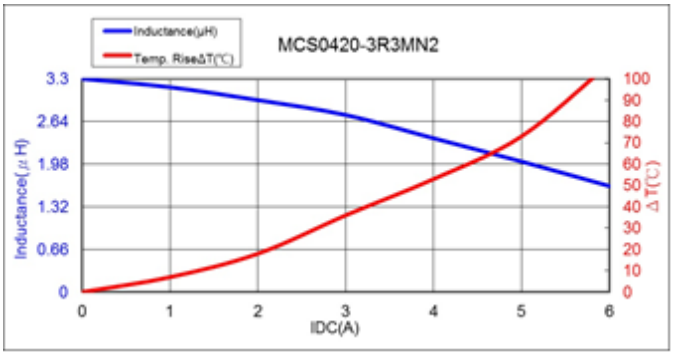
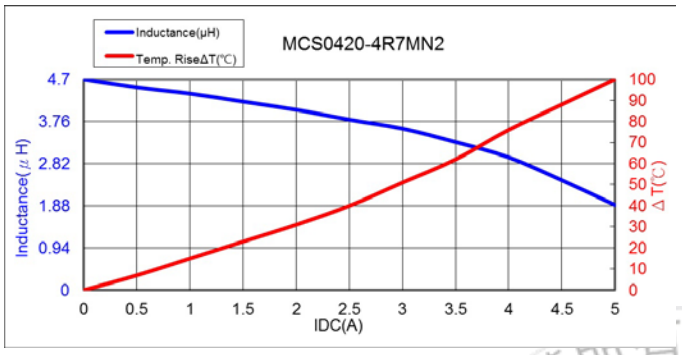
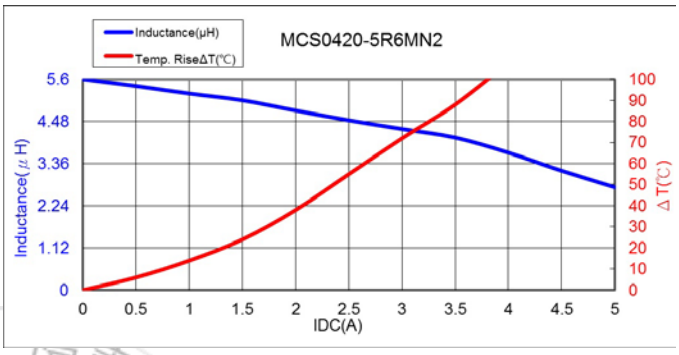
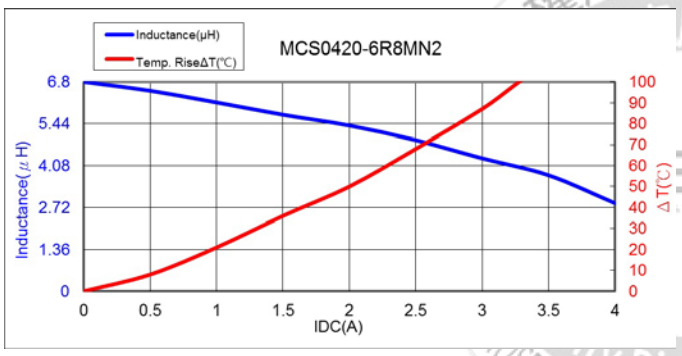
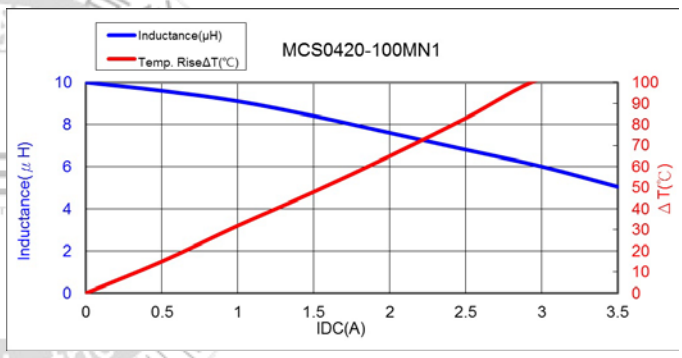








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